

Working of Single Lines — Revision Guide

Module 3 — Working of Single Lines | DC63 Driver Plan, Week 6 (22-26 June 2026)

Based on GERT8000-TW1 §33 and §34 (with a cross-reference at §42.3), and the RS521 Signalling Handbook principle of one train per section, cross-referenced to the classroom session notes (which name TW1 as the required source document for this topic).

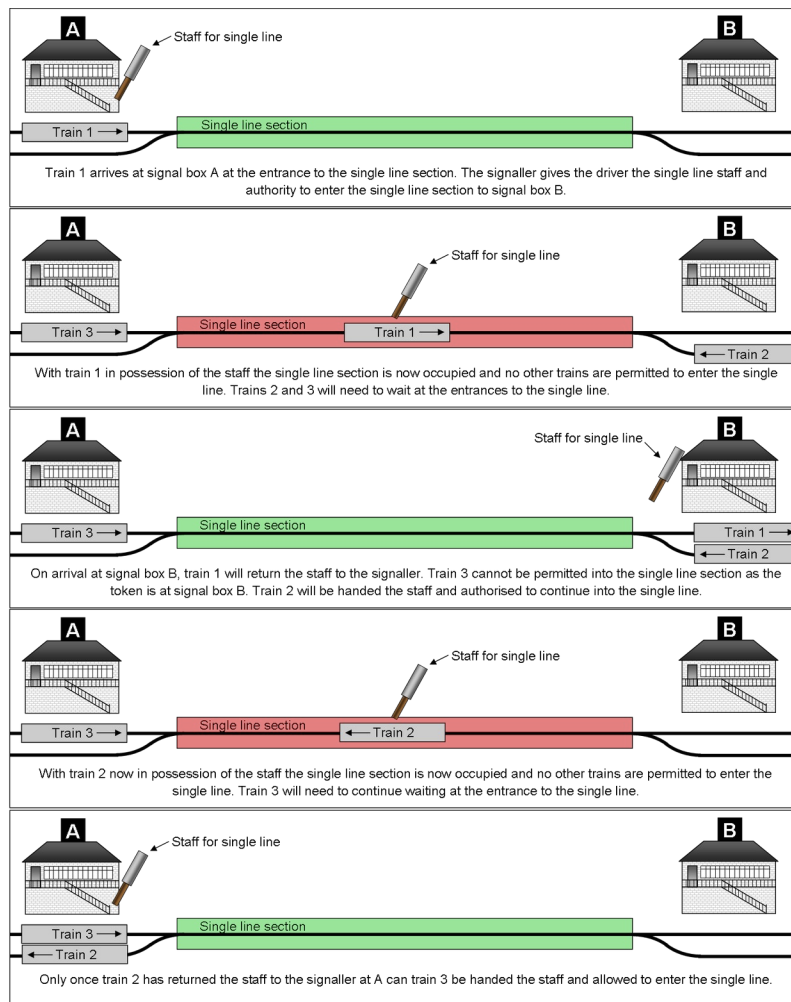
1. What Is Single Line Working?

Only one train at a time is allowed in a single-line section.

GERT8000-TW1 §34.1

Single lines use the same basic principle as absolute block signalling — one train per section — with the added requirement that trains can enter the section from either direction. A safe system of working is needed so that a train can only enter the section if it holds a recognised form of authority, often a physical object such as a token, staff, or ticket.

Classroom PPTX (Working of Single Lines), slides 3, 5



Staff-for-single-line working between signal box A and signal box B — only the driver holding the staff may enter the single line section.

Classroom PPTX (Working of Single Lines), slide 7

2. Getting the Token or Train Staff

You must always stop your train when you need to get, deliver, or exchange a token or train staff.

Before taking a train onto the single line, you must get the correct token or train staff for the section from the signaller or person authorised in the Sectional Appendix.

Where a no-signaller token instrument is provided, you must ask the signaller or authorised person to release the token.

If you are the driver at the leading end of the train, you must show the token or train staff to the driver of any other locomotive at the leading end before entering the single line section.

GERT8000-TW1 §34.2

3. When You Do Not Need the Token or Staff

You do not need the token or train staff if any of the following apply:

- The line is under possession.
- Working by pilot is in operation.
- Modified working arrangements are in operation.
- You are authorised to pass the section signal on an electric token line at danger for shunting purposes.
- Your train is to enter the single-line section as an assisting train.

GERT8000-TW1 §34.2

4. Handling the Token or Train Staff

You must keep the token or train staff with you in the cab from which the train is being driven until it is needed by a shunter.

If it has been given to the shunter for shunting purposes, you must not continue the journey until shunting is completed, the points are locked correctly for trains to pass on the single line, and the shunter has returned the token or train staff to you.

When the train reaches the end of the section, you must give the token or train staff to the signaller or authorised person, or — where a no-signaller token instrument is provided — place the token in the instrument or give it to the authorised person to do so.

GERT8000-TW1 §34.3

5. Train Failures and Assisting Trains

If your train has failed and an assisting train is to enter the section from a ground frame released by the token, the signaller will instruct you to take the token to the ground frame. On arrival you must contact the signaller, not place the token in the instrument, come to a clear understanding with the signaller about what is to be done, and hand the token to the driver of the assisting train.

If any portion of the train is left in the single-line section, you must tell the signaller before leaving the section, and keep the token or train staff until the whole train is clear of it.

If the signaller tells you the front portion of the train is to continue its journey, leaving the rear portion in the section, you must then give up the token or train staff.

GERT8000-TW1 §34.3

If the signaller tells you that, because of a failure of token instruments, trains will be run as if on a one-train working line with a train staff, you must handle the token as if it is a train staff, and not place it in any instrument.

On a no-signaller token line, you must not transfer the token from one train to another unless it has been passed through a token instrument, except when a train is entering to assist a portion left in the section, or when running under a token-instrument-failure arrangement as above.

GERT8000-TW1 §34.3

6. One-Train Working Without a Train Staff

You must not enter or foul the single-line section until the controlling signal is cleared, unless one of the following applies:

- The line is under possession.
- Working by pilot is in operation.
- Modified working arrangements are in operation.
- Your train is to enter the single-line section as an assisting train.

If any portion of the train is left in the single-line section, you must tell the signaller, and must not leave the section until you have done so.

GERT8000-TW1 §34.4

7. Wrong-Direction Working on a Single Line

If your train is to travel over the single line in the wrong direction, you must tell the guard. You must consider the effect on station working, releasing doors and passenger safety, and on protection arrangements under Rule Book module M1 (Dealing with a train accident or train evacuation).

GERT8000-TW1 §33.1

Where more than one running line is available and the single line working arrangements have not been published in the Weekly Operating Notice, you must still tell the guard if travelling in the wrong direction. Protection arrangements under module M1 must take into account the altered direction of train working.

GERT8000-TW1 §33.2

8. Returning to the Same End of the Section

If the train is returning to the same end of the section at which it entered, on a single or bi-directional line, you must ask the signaller for permission before the returning movement starts.

GERT8000-TW1 §42.3

9. AWS on Single & Bi-Directional Lines

On single and bi-directional lines, the AWS magnet is normally suppressed for movements it does not apply to, so the AWS will not operate for traffic moving in that direction.

Where the magnet is not suppressed, a cancelling indicator tells the driver that the AWS warning indication does not apply to trains travelling in that direction. The cancelling indicator is normally positioned 180 metres (approximately 200 yards) after the AWS magnet.

RS521 §5.15

On some bi-directional lines, AWS equipment is not provided at all in the opposite (wrong) direction. Signs mark the start of the relevant section of line and the end of the section where normal arrangements resume. However, for a temporary or emergency speed restriction, AWS will be provided in both directions.

RS521 §5.17

Suppressor magnets are used to suppress permanent magnets when they are not required to apply to a train movement — for example, magnets applicable to the opposite direction on a single or bi-directional line.

RS522 §1.2 / §1.5-1.6

TPWS — no single-line-specific provision

There is no separate single-line or bi-directional-line rule for TPWS in RS522. TPWS equipment is fitted at the relevant signals, speed restrictions, and buffer stops in the same way regardless of whether the line is single, double, or bi-directional.

RS522 §2

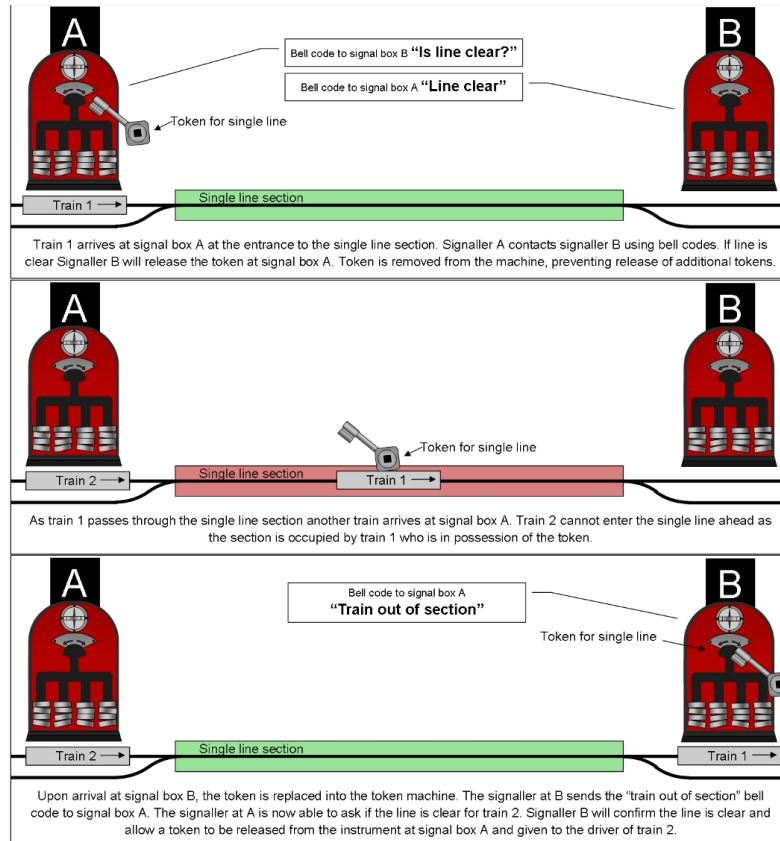
10. Evolution of Single Line Systems — Classroom Context

The token system was developed in 19th-century Britain to allow safe working of single-line railways. Originally called One Engine in Steam (OES), it is now known as One Train Working with Staff (OTS). The driver of any train entering the branch line must be in possession of the token, which prevents a collision with another train.

Classroom PPTX (Working of Single Lines), slide 6

Electric token working was introduced in the late 1800s once signal boxes could communicate by electric telegraph, overcoming the delays of the staff system. The diagram below shows the sequence: the driver stops at the section signal, the signaller exchanges bell codes ('Is line clear?', 'Line clear') with the signaller at the other end, and a token is only released from the instrument once both ends agree the section is clear.

Classroom PPTX (Working of Single Lines), slide 10

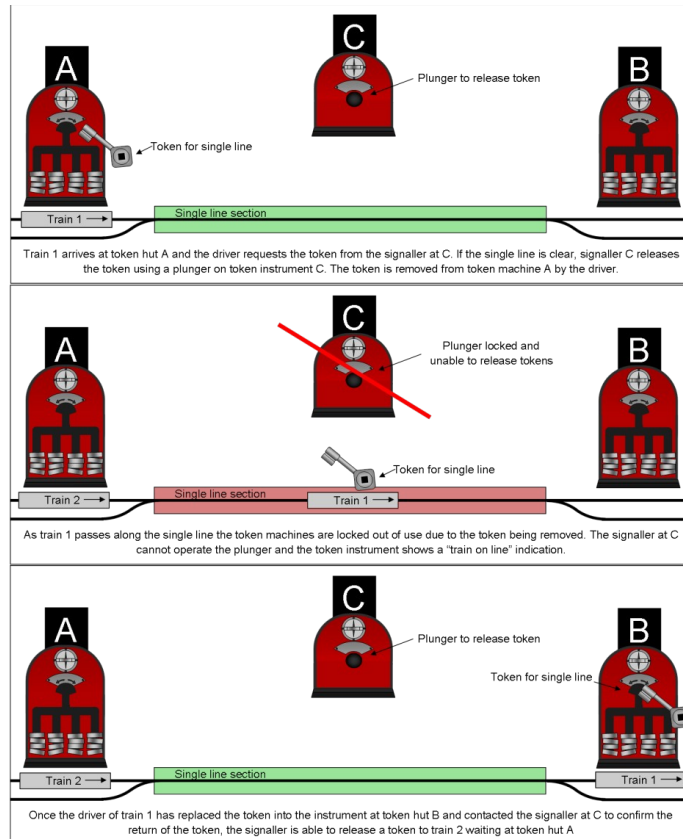


Electric Token Working — bell-code sequence between signal boxes A and B before a token is released for the single line section.

Classroom PPTX (Working of Single Lines), slide 10

No-signaller token (NST) working uses the same equipment and principles as electric token working, but no longer requires a signal box and signaller to be located at the entrance to each section — instead, a plunger-release token instrument allows the driver or an authorised person to obtain the token directly once the section is confirmed clear.

Classroom PPTX (Working of Single Lines), slide 11

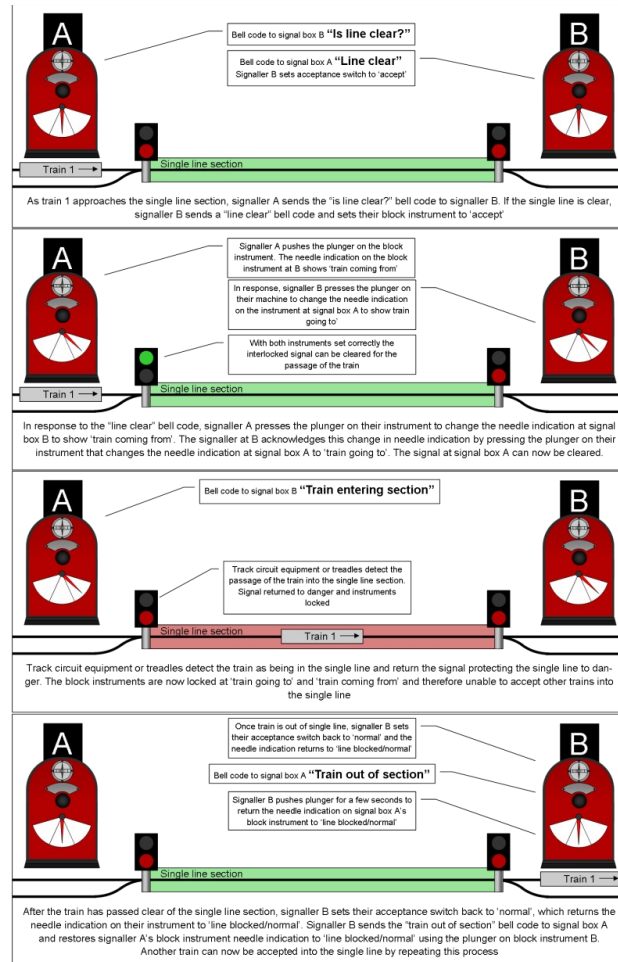


No Signaller Electric Token Working — token huts and plunger-release sequence where no signal box is provided at the section entrance.

Classroom PPTX (Working of Single Lines), slide 12

Tokenless block working went further, replacing the physical token with an interlocked starting signal that can only show a proceed indication when the section is clear and both signallers agree to the movement. The signaller at the opposite end operates an acceptance device, which then allows the signaller at the entrance end to clear the section signal — clearance of that signal is the only authority the driver needs to enter the single line.

Classroom PPTX (Working of Single Lines), slides 13-14



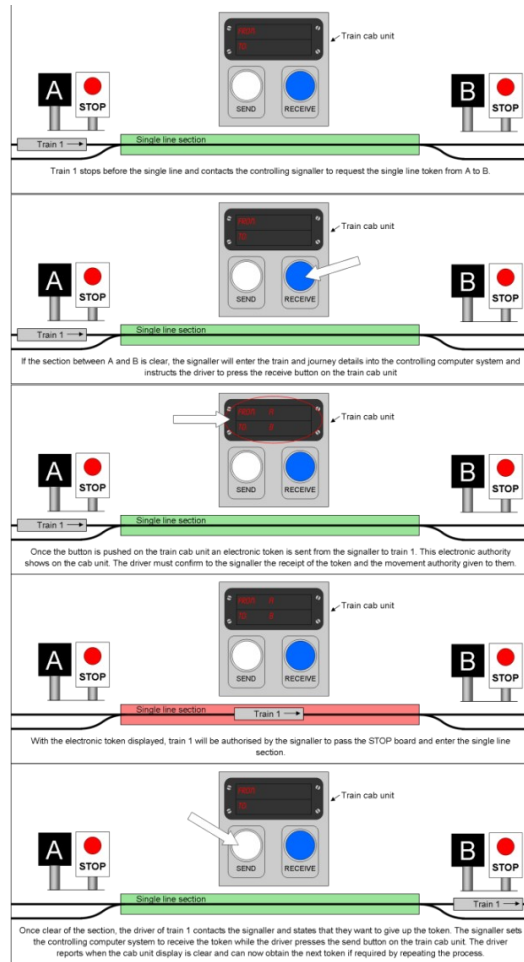
Tokenless Block Working — the section signal can only clear once the acceptance device confirms the section is clear and both signallers agree.

Classroom PPTX (Working of Single Lines), slide 14

Radio Electronic Token Block (RETB)

Radio Electronic Token Block dispenses with a physical token altogether. The driver contacts the signaller by radio from a cab unit and requests the token; once the signaller has entered the journey details and confirmed the section is clear, an electronic token is transmitted to the cab unit and displayed on it. The driver is then authorised to pass the stop board and enter the single line section, and gives the token back to the signaller by radio once clear at the other end.

Classroom PPTX (Working of Single Lines), slide 15



Radio Electronic Token Block (RETB) — the driver requests, receives, and surrenders the electronic token by radio via the cab unit, with no signal box at the section entrance.

Classroom PPTX (Working of Single Lines), slide 15

11. Bi-Directional Signalling & SIMBIDS — Classroom Context

A double (or multiple) track railway can be signalled to permit trains to run in either direction on any running line — termed bi-directional (or reversible) signalling — which may be 'full capacity' or 'reduced capacity'.

Simplified Bi-Directional Signalling (SIMBIDS) is a simplified, reduced-capacity form. No AWS equipment is provided for signals applying in the wrong direction (AWS for normal-direction signals is not suppressed), and wrong-direction running is limited to no more than 70 mph. Wrong-direction movements under SIMBIDS are only permitted during a possession of one line or during emergencies — not for the regular passage of booked trains.

Classroom PPTX (Working of Single Lines), slides 16-17

12. Case Study — Departing Without a Token

Classroom case study: a signaller forgot to give a driver the token for a single-line section before departure, and the driver departed without it. The Mobile Operations Manager (MOM) was sent to collect the token, and the train operating company authorised the driver to proceed to the next station, where they were relieved of duty. This illustrates why drivers must always positively confirm they hold the correct token or staff before entering a single-line section.

Classroom PPTX (Working of Single Lines), slide 18 — NR National Log item

Quick Reference

Item	Value	Reference
Trains allowed in a single-line section at once	One	TW1 §34.1
Original name for token working	One Engine in Steam (OES)	Classroom PPTX, slide 6
Modern name for token working	One Train Working with Staff (OTS)	Classroom PPTX, slide 6
Max wrong-direction speed under SIMBIDS	70 mph	Classroom PPTX, slide 17
Authority needed under tokenless block working	Cleared section signal only	Classroom PPTX, slide 14
Cancelling indicator position after AWS magnet	180 m (approx. 200 yards)	RS521 §5.15
How RETB authorises entry without a physical token	Electronic token sent to cab unit by radio	Classroom PPTX, slide 15

Revision Questions

1. How many trains are allowed in a single-line section at any one time?

Answer: Only one train at a time is allowed in a single-line section.

GERT8000-TW1 §34.1

2. What must you always do when you need to get, deliver, or exchange a token or train staff?

Answer: Always stop your train.

GERT8000-TW1 §34.2

3. Before taking a train onto the single line, what must you make sure you have?

Answer: The correct token or train staff for the section, obtained from the signaller or person authorised in the Sectional Appendix.

GERT8000-TW1 §34.2

4. If you are the leading driver of a train with another locomotive at the leading end, what must you do before entering the single line?

Answer: Show the token or train staff to the driver of that other locomotive.

GERT8000-TW1 §34.2

5. Name three situations where you do not need the token or train staff to enter a single-line section.

Answer: Any three of: the line is under possession; working by pilot is in operation; modified working arrangements are in operation; you are authorised to pass the section signal at danger for shunting on an electric token line; the train is entering as an assisting train.

GERT8000-TW1 §34.2

6. Where must the driver keep the token or train staff during the journey?

Answer: With them in the cab from which the train is being driven, until it is needed by a shunter.

GERT8000-TW1 §34.3

7. If the token or train staff has been given to a shunter, what three things must happen before the journey continues?

Answer: Shunting must be completed, the points must be locked correctly for trains to pass on the single line, and the shunter must have returned the token or train staff to the driver.

GERT8000-TW1 §34.3

8. What must the driver do with the token or train staff when the train reaches the end of the section?

Answer: Give it to the signaller or authorised person, or — where a no-signaller token instrument is provided — place it in the instrument or give it to the authorised person to do so.

GERT8000-TW1 §34.3

9. If a train has failed and an assisting train is to enter from a ground frame released by the token, what must the driver do on arrival at the ground frame?

Answer: Contact the signaller, not place the token in the instrument, come to a clear understanding with the signaller about what is to be done, and hand the token to the driver of the assisting train.

GERT8000-TW1 §34.3

10. If part of the train is left in the single-line section, what must the driver do regarding the token or staff?

Answer: Tell the signaller before leaving the section, and keep the token or train staff until the whole train is clear of the section.

GERT8000-TW1 §34.3

11. If token instruments fail and the signaller says the line will be run as a one-train working line with a train staff, how must the driver handle the token?

Answer: Handle the token as if it is a train staff, and not place it in any instrument.

GERT8000-TW1 §34.3

12. Under one-train working without a train staff, when may a driver not enter or foul the single-line section?

Answer: Not until the controlling signal is cleared, unless the line is under possession, working by pilot is in operation, modified working arrangements are in operation, or the train is entering as an assisting train.

GERT8000-TW1 §34.4

13. If part of the train is left in the section under one-train working without a train staff, what must the driver do?

Answer: Tell the signaller, and not leave the single-line section until they have done so.

GERT8000-TW1 §34.4

14. If your train is to travel over the single line in the wrong direction, who must you tell?

Answer: The guard.

GERT8000-TW1 §33.1

15. What must the driver and guard consider when travelling the single line in the wrong direction?

Answer: The effect on station working, releasing doors and passenger safety, and on protection arrangements under Rule Book module M1.

GERT8000-TW1 §33.1

16. If a train returns to the same end of a single or bi-directional line section at which it entered, what must the driver do first?

Answer: Ask the signaller for permission before the returning movement starts.

GERT8000-TW1 §42.3

17. On a single or bi-directional line, what normally happens to the AWS magnet for movements it does not apply to?

Answer: It is normally suppressed, so the AWS will not operate for movements travelling in that direction.

RS521 §5.15

18. Where the AWS magnet is not suppressed on a single or bi-directional line, what tells the driver the warning does not apply?

Answer: A cancelling indicator, normally positioned 180 metres (approximately 200 yards) after the AWS magnet.

RS521 §5.15

19. On some bi-directional lines, is AWS equipment always provided in the wrong direction?

Answer: No — on some bi-directional lines AWS equipment is not provided at all in the opposite direction, although AWS will still be provided in both directions for a temporary or emergency speed restriction.

RS521 §5.17

20. Is there a separate TPWS rule for single or bi-directional lines?

Answer: No — RS522 does not give TPWS a separate single-line or bi-directional-line provision; it is fitted at the relevant signals, speed restrictions, and buffer stops in the same way regardless of the type of line.

RS522 §2

21. What was the original name for token-based single line working, and what is it now called?

Answer: Originally One Engine in Steam (OES); now known as One Train Working with Staff (OTS).

Classroom PPTX (Working of Single Lines), slide 6

22. What technological advance allowed electric token working to be introduced?

Answer: Signal boxes being able to communicate with each other using the electric telegraph.

Classroom PPTX (Working of Single Lines), slide 9

23. How does no-signaller token (NST) working differ from electric token working?

Answer: It uses the same equipment and principles, but no longer requires a signal box and signaller to be located at the entrance to each single-line section.

Classroom PPTX (Working of Single Lines), slide 11

24. Under tokenless block working, what is the only authority a driver needs to enter the single line?

Answer: Clearance of the section signal — no physical token is required.

Classroom PPTX (Working of Single Lines), slide 14

25. What is the maximum speed for wrong-direction running under SIMBIDS?

Answer: No more than 70 mph.

Classroom PPTX (Working of Single Lines), slide 17

26. When are wrong-direction movements permitted under SIMBIDS conditions?

Answer: Only during possession of one line or during emergencies — not for the regular passage of booked trains.

Classroom PPTX (Working of Single Lines), slide 17

27. In the classroom case study, what went wrong at St Bees and how was it resolved?

Answer: The signaller forgot to give the driver the token before departure and the driver left without it; the Mobile Operations Manager retrieved the token, and the driver was authorised to proceed to the next station, where they were relieved of duty.

Classroom PPTX (Working of Single Lines), slide 18